Identities with Reciprocal Trigonometric Functions

Exam code: 9MA0

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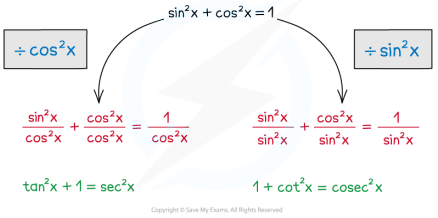
Identities with reciprocal trigonometric functions

What are the reciprocal trigonometric identities?

- There are two identities with **sec**, **cosec** and **cot** that you need to know and be able to use:
  - $\tan^2 x + 1 \equiv \sec^2 x$
  - $1 + \cot^2 x \equiv \operatorname{cosec}^2 x$
- These are not really 'new' identities – they can both be derived from $\sin^2 x + \cos^2 x \equiv 1$
  - To derive the identity for $\sec^2 x$ divide $\sin^2 x + \cos^2 x \equiv 1$ by $\cos^2 x$
  - To derive the identity for $\operatorname{cosec}^2 x$ divide $\sin^2 x + \cos^2 x \equiv 1$ by $\sin^2 x$



Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

Examiner Tips and Tricks

- These identities are not given in the exam formulae booklet – make sure you know how to find them.
- They will be needed to prove some trigonometric identities and solve some trigonometric equations.

Worked Example

Solve the equation $9\sec^2\theta - 11 = 3\tan\theta$,
in the interval $0 \leq \theta \leq 360°$.
Give your answers to 3 significant figures.

$(9\sec^2\theta - 9) - 2 = 3\tan\theta$

$\tan^2\theta + 1 = \sec^2\theta$
SO $\sec^2\theta - 1 = \tan^2\theta$

$9\tan^2\theta - 2 = 3\tan\theta$

$9\tan^2\theta - 3\tan\theta - 2 = 0$

THIS IS A QUADRATIC IN $\tan\theta$.
COMPARE $9y^2 - 3y - 2 = 0$

$(3\tan\theta - 2)(3\tan\theta + 1) = 0$

$\tan\theta = \frac{2}{3}$ OR $\tan\theta = -\frac{1}{3}$

USE $\tan^{-1}$ ON YOUR CALCULATOR TO FIND PRINCIPAL VALUES OF θ

IF $\tan\theta = \frac{2}{3}$

$\theta = 33.69$ OR $\theta = 180 + 33.69 = 213.69$

IF $\tan\theta = -\frac{1}{3}$

$\theta = -18.43$ OR $\theta = 360 - 18.43 = 341.57$

OR $= 180 - 18.43 = 161.57$

BUT THEN BE SURE TO FIND ALL OTHER VALUES OF θ IN THE REQUIRED RANGE

SO $= 33.7°, 162°, 214°, 342°$
(3 SIG. FIGS.)

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